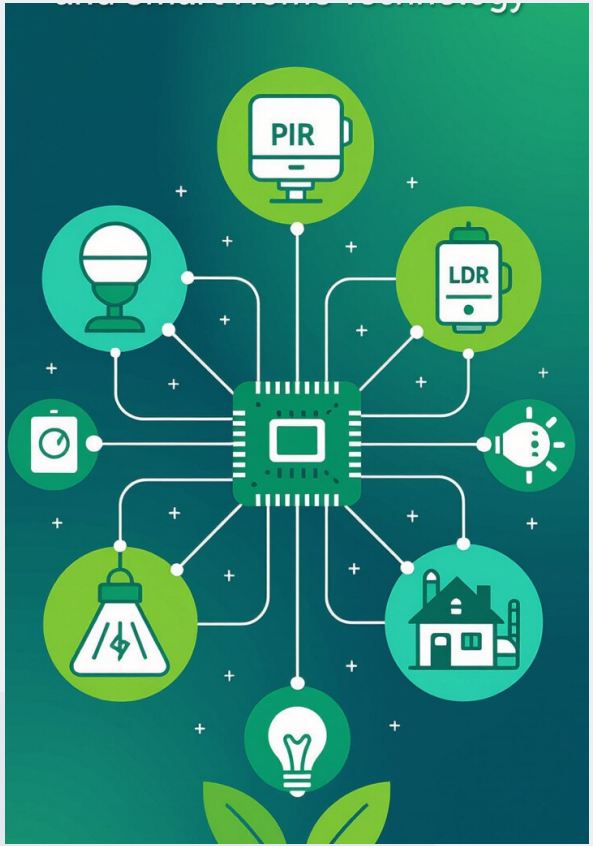
Title: Smart Light System using PIR Sensor and Light Sensor using Arduino Uno



BY - SARANSH PATEL
IBM_PBLP

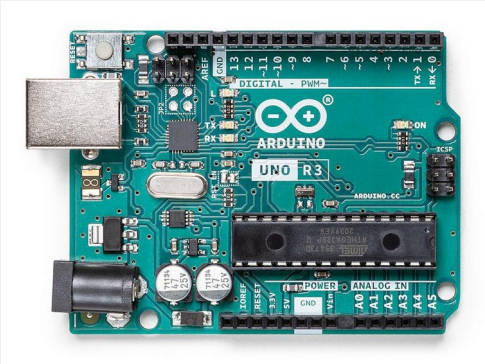
Objective of the Project

The goal of this project is to automate home lighting using PIR (Passive Infrared) and LDR (Light Dependent Resistor) sensors integrated with an Arduino UNO micro-controller.



- Aims to reduce energy consumption by activating lights only when needed.
- Integrates motion and ambient light detection for intelligent control.
- Showcases a practical application of embedded systems in smart home technology In TinkerCad.

Component List



- Arduino UNO: The microcontroller that processes sensor data and controls the relay.
- PIR Sensor : Detects motion by sensing changes in infrared radiation.
- LDR Sensor : Measures ambient light levels; resistance decreases as light increases.
- 1kΩ Resistor : Forms a voltage divider with the LDR to measure light intensity.
- Relay Module(SPDT) ： An electromechanical switch to control the AC bulb safely.
- AC Bulb : The light source being controlled.
- Power Supply : Provides necessary power to the Arduino and other components.
- Breadboard : Used for prototyping and connecting components.
- Wires : Connect all components together.

Name	Quantity	Component
U1	1	Arduino Uno R3
PIR1	1	PIR Sensor
K1	1	Relay SPDT
P1	1	5 , 5 Power Supply
L1	1	Light bulb
R1	1	Photoresistor
R2	1	1 kΩ Resistor

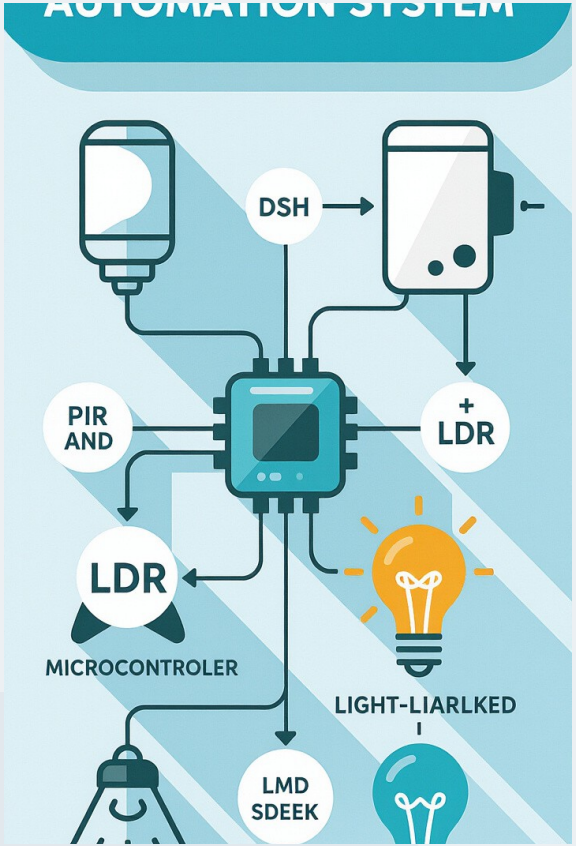
Working Principle

The system activates the AC bulb only when two conditions are met:

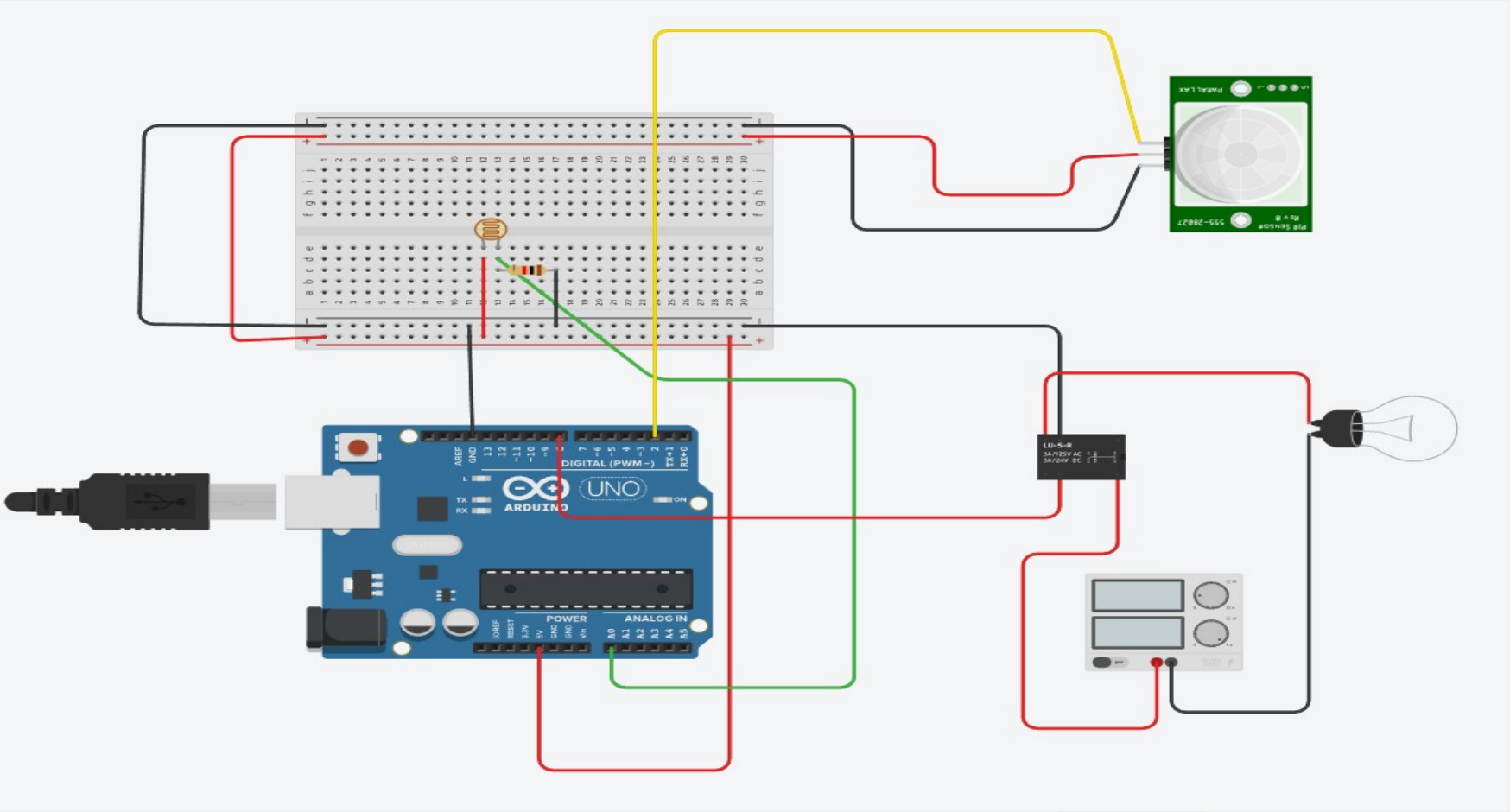
Motion Detection : The PIR sensor detects movement within its range.

Low Light : The LDR senses that the ambient light is below a certain threshold. If both conditions are true, the Arduino activates the relay, turning on the bulb.

Energy efficiency is achieved by preventing the bulb from turning on during daylight hours or when no one is present.



Circuit diagram

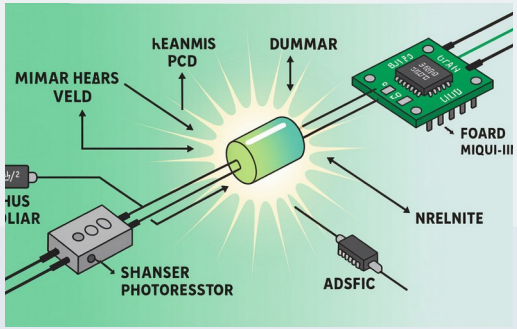


About Sensors and Relay



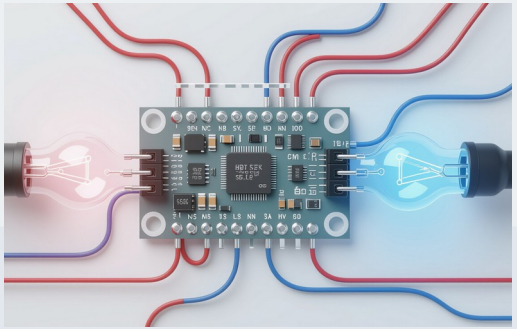
PIR Sensor

Detects motion by sensing changes in infrared radiation emitted by warm objects. Outputs a HIGH signal when motion is detected, and LOW otherwise. Adjustable sensitivity and time delay settings.



LDR

A photoresistor whose resistance varies based on the intensity of light. Used in a voltage divider circuit to convert light intensity into a voltage that the Arduino can read. Lower resistance in bright light and higher resistance in darkness.



Relay Module

An electromechanical switch that allows a low-voltage circuit (Arduino) to control a high-voltage circuit (AC bulb). Provides electrical isolation between the control circuit and the load circuit. Essential for safely controlling AC devices.

Code snippet

```
// C++ code

int LightSensor = 0;

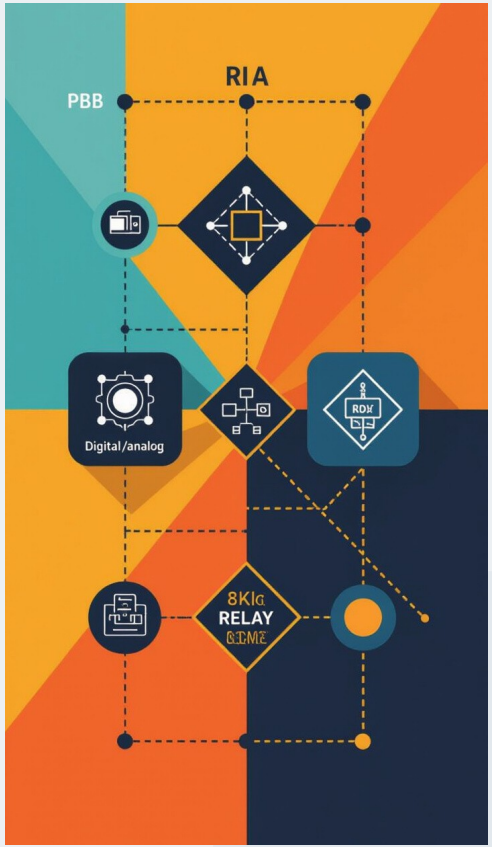
int LightSensorVal = 0;

int PIRSensorVal = 0;

int RelayOutputVal = 0;

void setup()
{
  pinMode(A0, INPUT);
  pinMode(2, INPUT);
  pinMode(8, OUTPUT);
  Serial.begin(9600);
}
```

```
void loop()
{
  LightSensorVal = analogRead(A0);
  PIRSensorVal = digitalRead(2);
  RelayOutputVal = 8;
  if (LightSensorVal < 600) {
    if (PIRSensorVal == HIGH) {
      digitalWrite(8, HIGH);
      delay(5000); // Wait for 5000 millisecond(s)
    } else {
      digitalWrite(8, LOW);
      delay(1000); // Wait for 1000 millisecond(s)
    }
  } else {
    digitalWrite(8, LOW);
    Serial.println(LightSensorVal);
    delay(100); // Wait for 100 millisecond(s)
  }
}
```



Arduino Code Explanation

The Arduino code performs the following main functions:

- Sensor Reading: Reads the digital value from the PIR sensor and the analog value from the LDR.
- Pin Mode Setup: Configures the digital pins connected to the PIR sensor and relay module as inputs and outputs, respectively.
- Conditional Logic: Implements an if statement to check if motion is detected and the light level is low.
- Relay Control: Sets the digital output pin connected to the relay module HIGH or LOW to turn the bulb ON or OFF, respectively.

Code Review - Highlights

- Motion Detection: Uses digitalRead(PIR_PIN) to check for motion; a HIGH value indicates motion.
- Light Sensing: Uses analogRead(LDR_PIN) to read the analog value from the LDR. Higher analog values indicate darker conditions.
- Conditional Bulb Activation : The core logic is in the if statement: c++ if (motionDetected && isDark) { digitalWrite(RELAY_PIN, HIGH); // Turn on the bulb } else
- Safety and Energy Efficiency: The system ensures the bulb only turns on when needed, reducing energy waste and improving safety by providing light when motion is detected in the dark.



Bulb turns on - Both Condition met

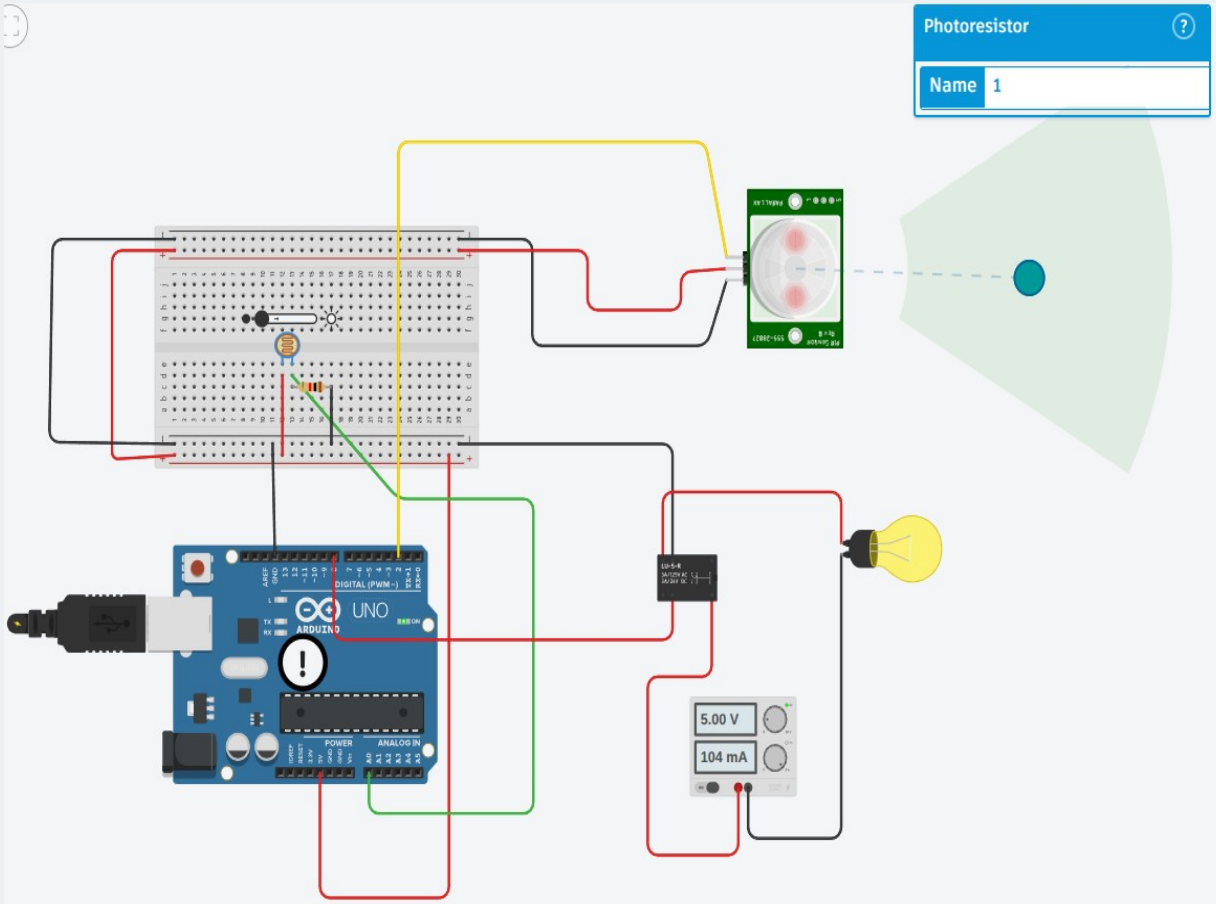
Conditions Met (from code) :

```
if (LightSensorVal < 600)  {
  if (PIRSensorVal == HIGH)  {
    digitalWrite(8, HIGH);    // Turn ON bulb
  }
}
```

Explanation:

- LightSensorVal < 600: It's dark (night time), no value print on serial monitor .
- PIRSensorVal == HIGH: Motion is detected.

Result: Bulb turns ON and remains ON for 5 seconds (delay(5000)).



1 (Arduino Uno R3)

```
// C++ code
//
int LightSensor = 0;

int LightSensorVal = 0;

int PIRSensorVal = 0;

int RelayOutputVal = 0;

void setup()
{
  pinMode(A0, INPUT);
  pinMode(2, INPUT);
  pinMode(8, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  LightSensorVal = analogRead(A0);
  PIRSensorVal = digitalRead(2);
  RelayOutputVal = 8;
  if (LightSensorVal < 600) {
    if (PIRSensorVal == HIGH) {
      digitalWrite(8, HIGH);
      delay(5000); // Wait for 5000 millisecond(s)
    } else {
      digitalWrite(8, LOW);
      delay(1000); // Wait for 1000 millisecond(s)
    }
  } else {
    digitalWrite(8, LOW);
    Serial.println(LightSensorVal);
    delay(100); // Wait for 100 millisecond(s)
  }
}
```

Serial Monitor

Bulb Remains OFF – Daylight Present

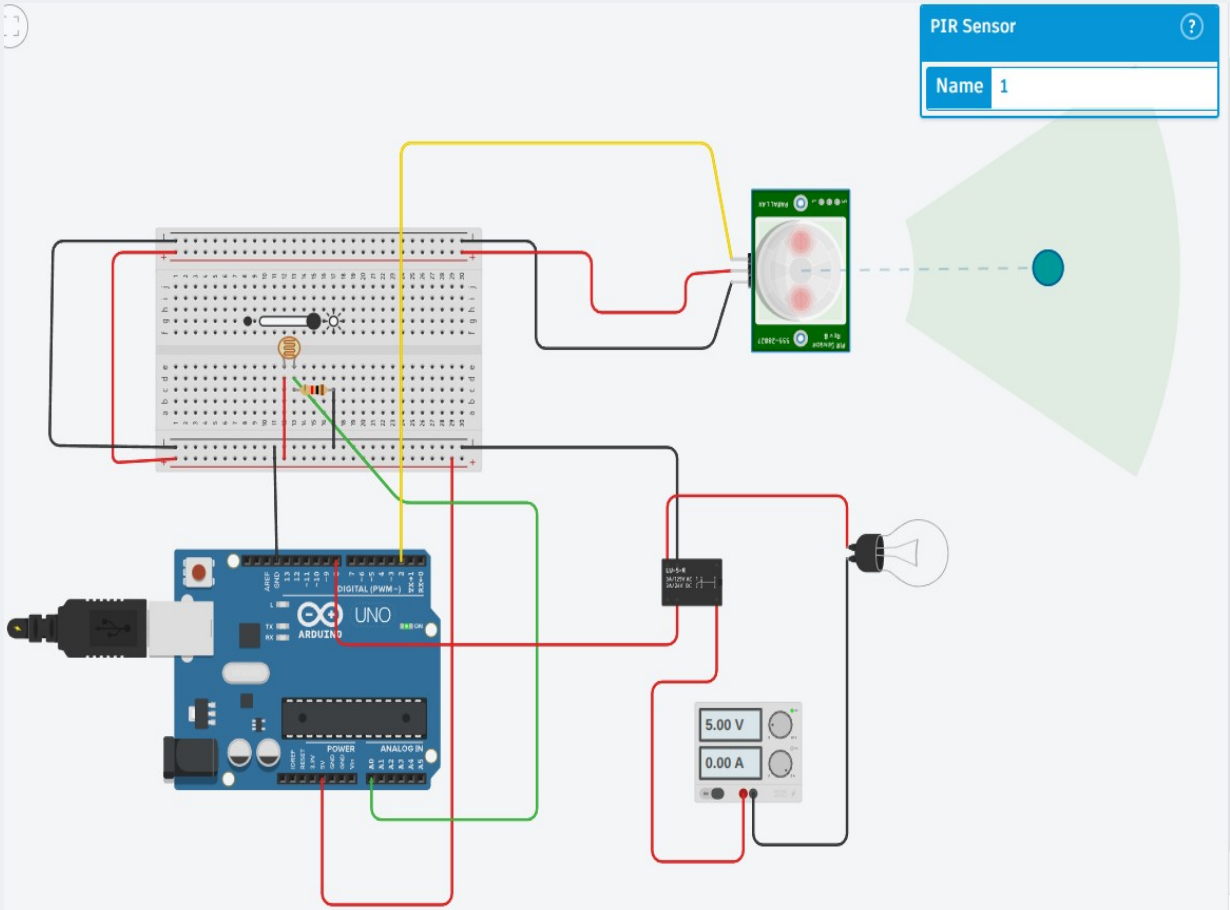
Conditions Triggered (from code):

```
if (LightSensorVal >= 600) {
  digitalWrite(8, LOW); // Keep bulb OFF
}
```

Explanation:

- LightSensorVal >= 600: It's bright (day time), value printed on serial monitor (675 >600) .
- Regardless of PIR sensor state, bulb will stay OFF

Result: Even if motion is detected, bulb remains OFF to save energy.



1 (Arduino Uno R3)

```
// C++ code
//
int LightSensor = 0;

int LightSensorVal = 0;

int PIRSensorVal = 0;

int RelayOutputVal = 0;

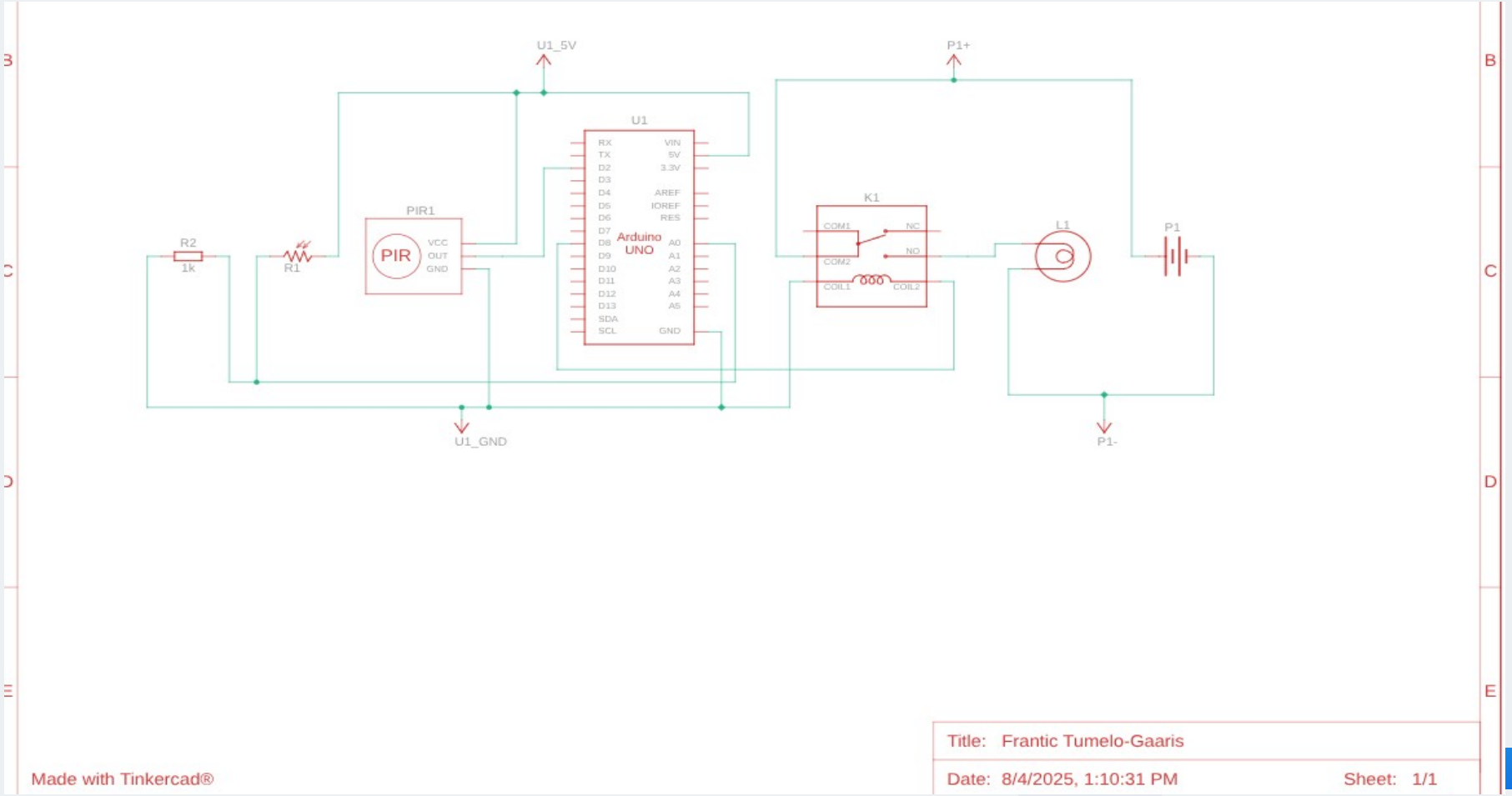
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  pinMode(8, OUTPUT);
  Serial.begin(9600);
}

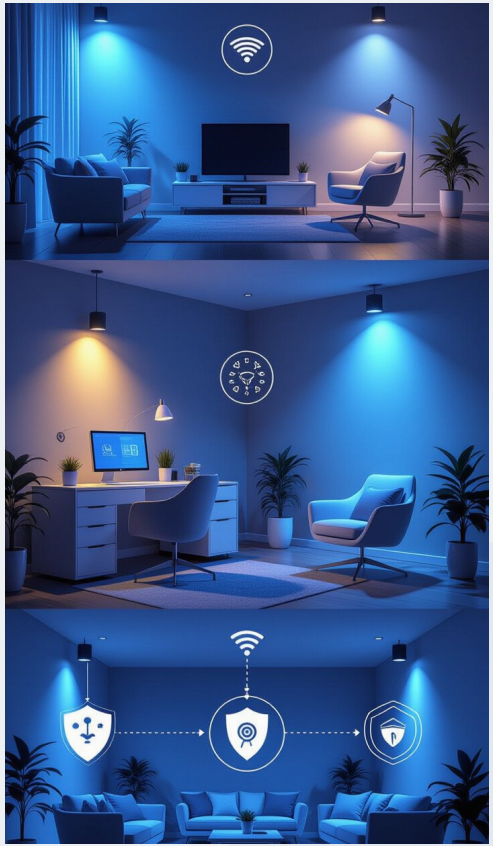
void loop()
{
  LightSensorVal = analogRead(A0);
  PIRSensorVal = digitalRead(2);
  RelayOutputVal = 8;
  if (LightSensorVal < 600) {
    if (PIRSensorVal == HIGH) {
      digitalWrite(8, HIGH);
      delay(5000); // Wait for 5000 millisecond(s)
    } else {
      digitalWrite(8, LOW);
      delay(1000); // Wait for 1000 millisecond(s)
    }
  } else {
    digitalWrite(8, LOW);
    Serial.println(LightSensorVal);
    delay(100); // Wait for 100 millisecond(s)
  }
}
```

Serial Monitor

```
675
675
675
675
675
```

Schematic view





Real-World Applications

- Home Lighting Automation: Automatically control indoor and outdoor lights based on occupancy and ambient light.
- Office Energy Saving: Reduce electricity bills by ensuring lights are only on when needed in office spaces.
- Smart Building Integration: Integrate with building management systems for centralized control and monitoring of lighting.
- Security Systems: Use motion-activated lighting as a deterrent for intruders.

Benefits of the System

- Energy-Efficient: Reduces energy consumption by only activating lights when necessary.
- Cost-Effective: Low component cost and potential savings on electricity bills.
- Scalable: Easily expandable to control multiple lights and integrate with other smart home devices.
- Improves User Comfort and Safety: Provides convenient and automatic lighting when and where it's needed, enhancing safety.
- Demonstrates IoT Principles: Provides a hands-on example of connecting sensors and actuators for automated control.





Conclusion

This project successfully demonstrates the implementation of a smart home lighting system using Arduino, PIR, and LDR sensors.

- Provides a practical example of automation principles.
- Highlights the potential for energy savings and improved user experience.
- Offers a foundation for more advanced smart home applications.
- This project illustrates a real-world relevant application of IoT and embedded systems concepts using Arduino.